Sunset Views

AlgoExpert · Medium · Stacks

The problem

- Buildings face EAST or WEST; return ascending indices of buildings that can see the sunset (strictly taller than everything in front of them in that direction).

Intuition

- A building sees the sunset if it's taller than everything between it and the sun.

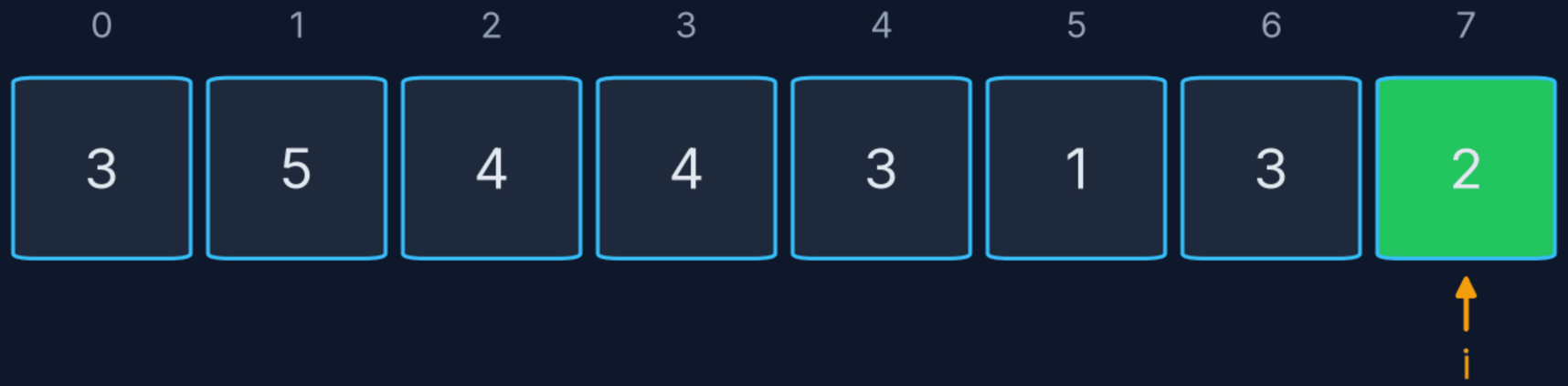
Brute force vs optimal

- For each building check all ahead → $O(n^2)$
- Scan from the sunset side tracking the max → $O(n)$
- Or a stack scanning the other way

Approach

- 1. Scan from the sunset side
- 2. Keep the running max
- 3. Buildings taller than the max can see
- 4. Return indices in order

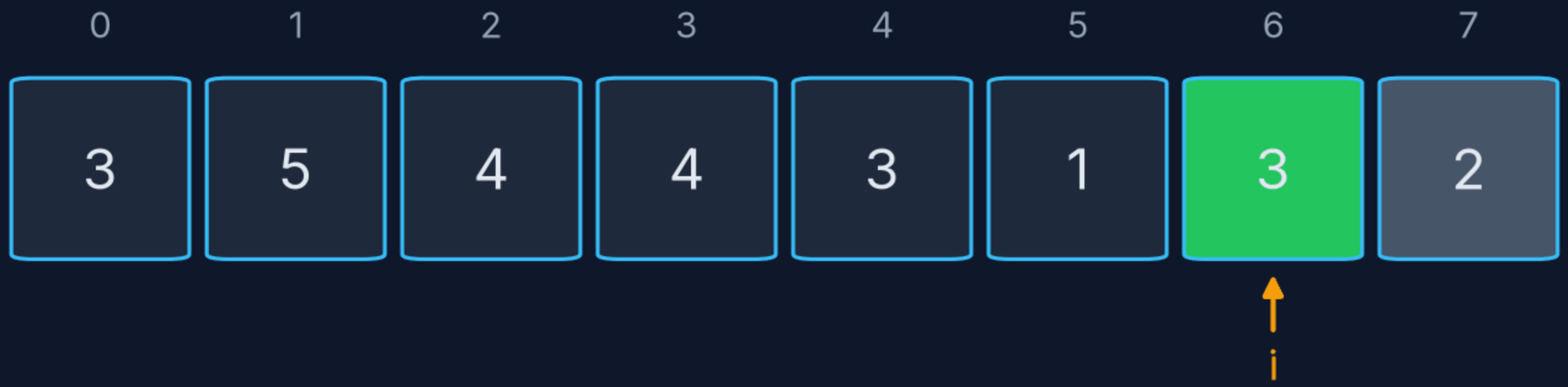
Worked example




```
tallest to the east so far = 2
```

Building 7 (height 2): sees sunset

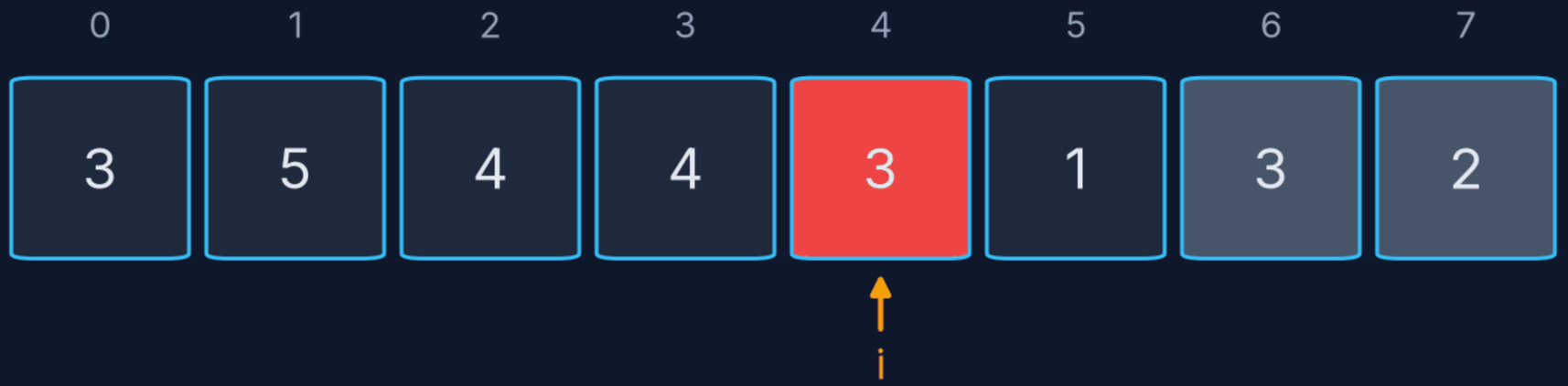
Worked example




```
tallest to the east so far = 3
```

Building 6 (height 3): sees sunset

Worked example




```
tallest to the east so far = 3
```

Building 4 (height 3): blocked

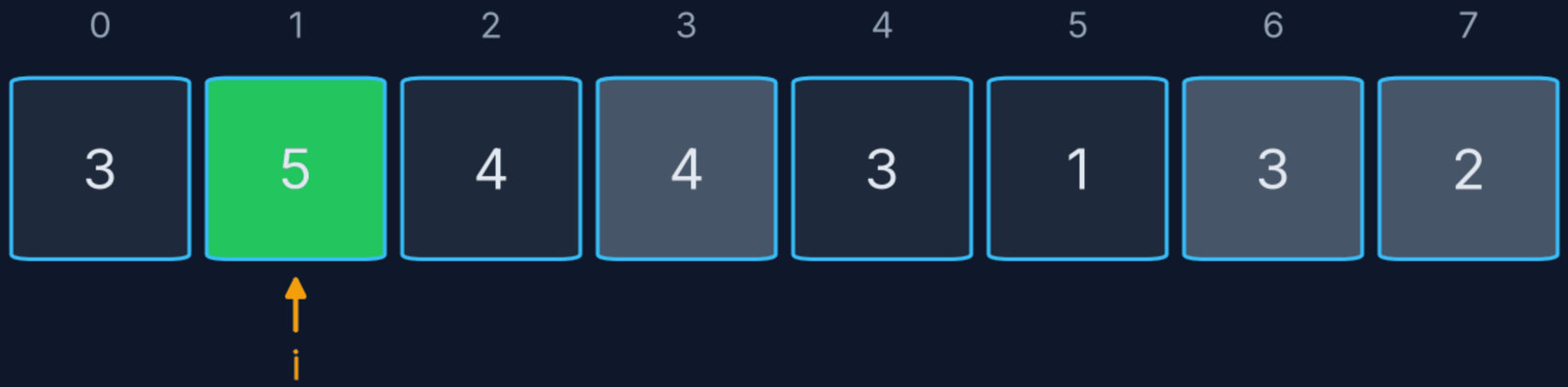
Worked example




```
tallest to the east so far = 4
```

Building 3 (height 4): sees sunset

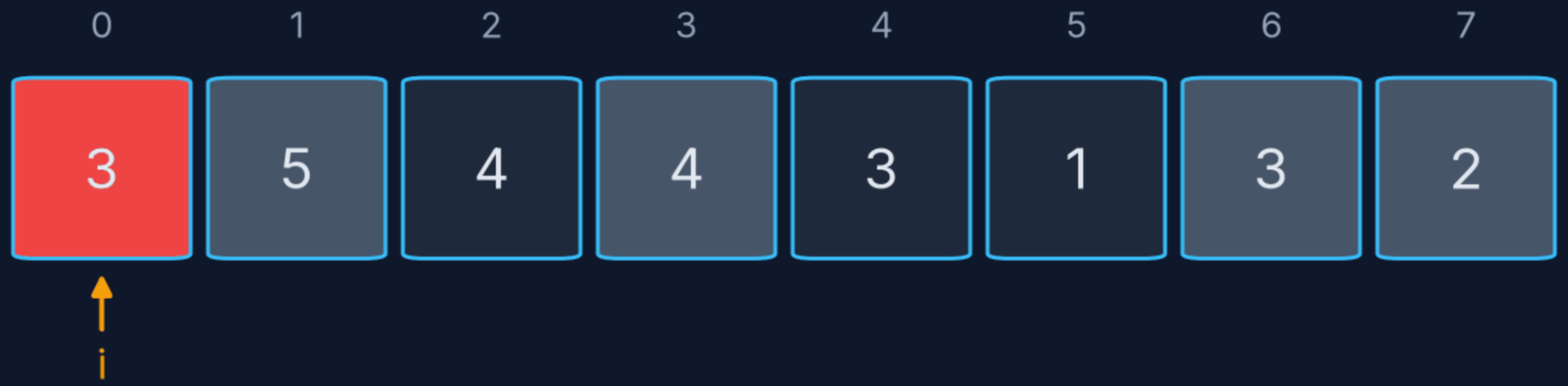
Worked example




```
tallest to the east so far = 5
```

Building 1 (height 5): sees sunset

Worked example




```
tallest to the east so far = 5
```

Building 0 (height 3): blocked

Java solution

```java
public class SunsetViews {
    public static List<Integer> sunsetViews(int[] heights, String direction) {
        List<Integer> out = new ArrayList<>();
        boolean east = direction.equals("EAST");
        int max = Integer.MIN_VALUE;
        for (int k = 0; k < heights.length; k++) {
            int i = east ? heights.length - 1 - k : k;
            if (heights[i] > max) { out.add(i); max = heights[i]; }
        }
        if (east) Collections.reverse(out);
        return out;
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Edge cases & pitfalls

- Equal heights block
- Return indices in ascending order
- Direction EAST vs WEST

Interview tips & follow-ups

- Mention the stack version

Recap

- A building sees the sunset if it's taller than everything between it and the sun.
- Complexity: O(n)